

Nicole D. Jackson, Ph.D.

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Professional Summary

- Climate risk and infrastructure analytics leader with 10+ years applying machine learning, statistics, and systems analysis to extreme-weather impacts on energy, water, and food systems.
- Principal Member of the Technical Staff at Sandia National Laboratories, translating climate, hazard, operational, and asset-level data into decision-ready risk insights.
- Technical focus: spatio-temporal modeling, probabilistic prediction, tail risk, stochastic scenario generation, uncertainty quantification, and applied AI.
- 44 peer-reviewed publications, 1 pending patent, \$3.9M managed of \$15.7M in secured funding, IEEE Resilience Week 2025 Best Paper, and National Academies committee service.

Professional Experience

Sandia National Laboratories, Albuquerque, NM

Principal Member of the Technical Staff

Senior Member of the Technical Staff

Postdoctoral Appointee

11/2019–Present

04/2025–Present

07/2021–04/2025

11/2019–07/2021

Climate Risk and Infrastructure Impact Modeling

- Built spatio-temporal Markov random field models to estimate outage probabilities following compound wildfire and flood events in California and Texas.
- Applied outage probabilities in a Texas capacity expansion use case to evaluate grid reliability under extreme-event scenarios.
- Fused outage records, storm reports, and gridded reanalysis across 3,000+ U.S. counties to characterize outage drivers.
- Quantified solar generation impacts across 186 sites in 16 states using reanalysis, station observations, and storm-event data.
- Measured solar generation impacts from hurricanes (−12.6%), snow (−54.5%), and storms (+1.1%) to support renewable-energy risk assessment.
- Developed wildfire-smoke solar loss model ($R^2 = 0.91$), isolating average smoke-driven derates of 8.3% across 53 sites.
- Modeled drought-driven water allocation trade-offs using multi-decadal hydrologic data across Texas basins.

AI, Statistical Modeling, and Scenario Generation

- Principal Investigator, EPIC PLANN (\$1.4M): developing graph-based reservoir computing for sub-seasonal, watershed-scale drought prediction across western U.S. basins using the 4-km CONUS404 reanalysis.
- Demonstrated preliminary skillful drought forecast horizons of ~13 weeks in select western U.S. basins.
- Co-invented Graph Reservoir Networks for prediction of spatiotemporal systems (patent 19/550,784 pending).
- Developed LSTM emulator of monthly water allocation behavior ($MSE = 0.034$) for rapid hydrologic-extreme scenario evaluation.
- Generated 21,000 climate-informed streamflow sequences using Bayesian HMMs to evaluate water allocations under extremes.
- Co-developed 50,000 GAN-based synthetic extreme weather scenarios to stress-test power-system tail behavior.
- Applied Random Forest and LIME methods in R to identify weather, operational, and metadata variables associated with low solar performance.

Research Leadership and Translation

- Directed 8 multi-hazard modeling projects for DOE programs, utilities, national laboratories, universities, and industry partners.
- Led 2–7 person cross-disciplinary teams to deliver modeling milestones, budgets, technical reports, and stakeholder-facing briefings.

- Managed \$3.9M in research funding within a \$15.7M secured portfolio spanning climate risk, grid resilience, energy-water systems, and applied AI.
- Processed 50,000+ O&M records from 800+ solar sites in 24 states and integrated them with weather and storm-event data.
- Documented model inputs, assumptions, uncertainty, and data lineage to support reproducibility, review, and stakeholder trust.
- Mentored 4 postdocs, 4 graduate students, and 2 undergraduates in hazard modeling and data-driven risk analysis, plus 29 senior design students.

University of Illinois at Urbana-Champaign, Graduate Research Assistant

2014–2019

- Built probabilistic global maps of crop-specific areas and a framework quantifying crop exposure to temperature and moisture extremes for 17 crops over 50+ years.
- Conducted network analysis of global food trade dynamics to evaluate how climate shocks in production regions propagate through agricultural systems.
- Produced 4 journal papers and 2 public datasets from interdisciplinary research linking climate variability, water resources, agriculture, and trade.

Technical Skills

- **R** primary for statistical modeling: tidyverse, caret, lime, extRemes, shiny.
- **Python**: pandas, geopandas, NumPy, SciPy, statsmodels, NetworkX, PyTorch, ReservoirPy, SHAP, git.
- **AI / Machine Learning**: Random Forests, LSTM, reservoir computing, graph reservoir networks, Bayesian HMMs, Markov random fields, explainable AI.
- **Statistical methods**: extreme value analysis, tail risk, spatio-temporal modeling, stochastic scenario generation, probabilistic prediction, uncertainty quantification.
- **Data**: gridded reanalysis and climate model output, station observations, storm-event records, outage data, asset-level performance data, hydrologic records, operational logs.
- **Domains**: climate risk, critical infrastructure resilience, renewable energy, electric grid reliability, drought and hydrologic prediction, energy-water systems, agricultural climate exposure.

Selected Recognition

- **Best Paper Award**, IEEE Resilience Week (highest reviewer score among all accepted papers) 2025
- **Member**, National Academies Committee on “Attribution of Extreme Weather and Climate Events and their Impacts” 2024–2026
- **Most Promising Engineer in Industry Award**, Black Engineer of the Year Awards 2024
- **MGB-SIAM Early Career Fellow** (1 of 10, inaugural class) 2022–2024

Selected Applied Outputs

- Co-invented Graph Reservoir Networks for prediction of spatiotemporal systems; patent 19/550,784 pending.
- Developed award-winning Markov random field approach for modeling weather-related compound threats to power systems.
- Contributed to open-source and public-data outputs supporting empirical analysis of photovoltaic operations and weather impacts.

Education

- **Ph.D., Civil Engineering**, University of Illinois at Urbana-Champaign 2020
- **M.S., Materials Science & Engineering**, University of Illinois at Urbana-Champaign 2014
- **M.S., Engineering Mechanics**, Virginia Tech 2008
- **B.S., Civil Engineering**, University of Illinois at Urbana-Champaign 2005